Probability Theory: Fundamentals and Applications

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1. Definition of Probability

Basic Probability Definition: Probability is a measure of uncertainty of events in a random experiment. It follows the axiomatic approach formulated by Russian Mathematician A.N. Kolmogorov (1903-1987), treating probability as a function of outcomes of the experiment.

Probability assigns numerical values between 0 and 1 to events, where: - **0** represents impossibility - **1** represents certainty - Values between 0 and 1 represent varying degrees of likelihood

Sample Space and Events: - **Sample Space (S)**: The set of all possible outcomes of a random experiment - **Event (E)**: A subset of the

sample space - **Probability Function**: P(E) assigns a number between 0 and 1 to each event E

2. Basic Probability Rules

2.1 The Three Fundamental Axioms

1. **Non-negativity**: P(E) ≥ 0 for any event E
2. **Normalization**: P(S) = 1 for the sample space S
3. **Additivity**: For mutually exclusive events, P(E₁ ∪ E₂) = P(E₁) + P(E₂)

2.2 Essential Calculation Rules

Addition Rule (Union of Events):

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Multiplication Rule (Intersection of Events):

$$P(A \cap B) = P(A) \times P(B|A) = P(B) \times P(A|B)$$

Complement Rule:

$$P(A') = 1 - P(A)$$

where A' represents the complement of event A.

2.3 Independence

Events A and B are **independent** if:

$$P(A \cap B) = P(A) \times P(B)$$

Or equivalently, if P(A|B) = P(A) and P(B|A) = P(B).

3. Examples with Dice and Coins

3.1 Single Die Roll

Sample Space: S = {1, 2, 3, 4, 5, 6}

Each outcome has equal probability: P(each outcome) = 1/6

Examples: - P(even number) = P({2, 4, 6}) = 3/6 = 1/2 - P(number > 4) = P({5, 6}) = 2/6 = 1/3 - P(rolling a 7) = 0 (impossible event)

3.2 Fair Coin Flip

Sample Space: S = {H, T} where H = Heads, T = Tails

Each outcome has equal probability: P(H) = P(T) = 1/2

Multiple Coin Flips: For two coin tosses: S = {HH, HT, TH, TT} - Each outcome has probability: 1/2 × 1/2 = 1/4 - P(exactly one head) = P({HT, TH}) = 2/4 = 1/2 - P(at least one head) = P({HH, HT, TH}) = 3/4

3.3 Random Variables Example

Coin Toss Experiment: Tossing a coin two times Sample Space: S = {HH, HT, TH, TT}

Random Variable X (number of heads): - X(HH) = 2, P(X = 2) = 1/4 - X(HT) = 1, X(TH) = 1, P(X = 1) = 2/4 = 1/2 - X(TT) = 0, P(X = 0) = 1/4

Random Variable Y (heads minus tails): - Y(HH) = 2, P(Y = 2) = 1/4 - Y(HT) = 0, Y(TH) = 0, P(Y = 0) = 2/4 = 1/2 - Y(TT) = -2, P(Y = -2) = 1/4

3.4 Two Dice Example

Sample Space: 36 equally likely outcomes (6 × 6)

Examples: - P(sum = 7) = 6/36 = 1/6 - Favorable outcomes: (1,6), (2,5), (3,4), (4,3), (5,2), (6,1) - P(sum = 2) = 1/36 - Favorable outcome: (1,1) - P(sum ≥ 11) = P(sum = 11 or 12) = 3/36 = 1/12 - Favorable outcomes: (5,6), (6,5), (6,6)

4. Conditional Probability

4.1 Definition

Conditional Probability answers: "What's the probability of event E, given that event F has occurred?"

If E and F are two events associated with the same sample space, the conditional probability of event E given that F has occurred is:

$$P(E|F) = \frac{P(E \cap F)}{P(F)}$$

provided P(F) ≠ 0.

4.2 Conceptual Understanding

Conditional probability addresses how the occurrence of one event affects the probability of another event. When we know that event F has occurred, we restrict our attention to only those outcomes where F happens, and calculate the probability of E within this restricted space.

4.3 Card Drawing Example

Drawing from a standard deck without replacement:

First card drawn is an Ace: - Original probability: P(Ace) = 4/52 = 1/13 - After drawing one Ace: 3 Aces remain out of 51 cards - P(second card is Ace | first card is Ace) = 3/51 = 1/17

This demonstrates how conditional probability differs from the original probability because the first draw changed the deck's composition.

4.4 Independence and Conditional Probability

Independent Events: If events A and B are independent, then: - P(A|B) = P(A) - P(B|A) = P(B) - P(A ∩ B) = P(A) × P(B)

Dependent Events: If events are dependent, then: - P(A|B) ≠ P(A) - Conditional probability becomes essential for calculations

4.5 Bayes' Theorem

An important application of conditional probability:

$$P(A|B) = \frac{P(B|A) \times P(A)}{P(B)}$$

This powerful tool allows us to update probabilities based on new evidence and is fundamental in many practical applications.

5. Problem-Solving Strategy

5.1 Systematic Approach

1. **Identify the sample space** and all possible outcomes
2. **Determine if outcomes are equally likely** or assign appropriate probabilities
3. **Define events clearly** using set notation
4. **Apply appropriate rules** (addition, multiplication, conditional probability)
5. **Verify results** make intuitive sense

5.2 Common Applications

Medical Testing: - P(disease | positive test) depends on both test accuracy and disease prevalence - Demonstrates the practical importance of conditional probability

Quality Control: - Probability of defective items given certain manufacturing conditions - Uses conditional probability to improve production processes

Weather Prediction: - Probability of rain given current atmospheric conditions - Combines multiple conditional probabilities for accurate forecasting

6. Key Takeaways

1. **Probability quantifies uncertainty** using values between 0 and 1
2. **Basic rules** (addition, multiplication, complement) are fundamental tools
3. **Conditional probability** is essential when events influence each other
4. **Random variables** connect outcomes to numerical values
5. **Practice with concrete examples** (dice, coins, cards) builds intuition
6. **Real-world applications** demonstrate the practical power of probability theory

Understanding these fundamentals provides a solid foundation for more advanced topics in probability and statistics, including probability distributions, hypothesis testing, and statistical inference.